Physics-Based Monte Carlo Simulation and a Hybrid Machine Engine for Landslide Risk Index Assessment

Conceptualisation, mathematical formulation, model behaviour, machine learning engine, and acceptance protocol

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Abstract

Landslide susceptibility mapping in Indonesia commonly relies on the Analytical Hierarchy Process (AHP), whose factor weights are elicited from experts and cannot be converted into a probability. This report formulates a physics-based alternative and audits it. Six field variables are mapped onto literature-based distributions of physical parameters, propagated through infinite-slope and Bishop limit-equilibrium equations by Monte Carlo simulation, and summarised as a probability of failure P_f; exact Sobol' indices over the 5^6 score space quantify the influence of each variable, and gradient, Jacobian and Hessian matrices describe parameter interaction. A hybrid engine, gradient boosting constrained to be monotone in every physical input, was validated on 575 published slope case histories with grouped cross-validation. On all cases (the pre-specified analysis) it reached AUC = 0.860, against 0.731 for logistic regression, 0.645 for physics alone and 0.581 for an equal-weight score. An audit (Section 7) corrected earlier, more favourable figures and found three limits that must be stated plainly: raw Monte Carlo P_f is not a calibrated probability on real slopes (ECE = 0.33); the score-based risk index used in the field has not been validated against observed landslides; and the variance-based weights depend strongly on how score classes are mapped to parameters. The engine passes 16 of 19 acceptance criteria (4 of them design checks), fails 1 and has 2 pending. It is a research and screening tool, not yet a basis for evacuation decisions.

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References

Notation. c' effective cohesion (kPa); φ' effective friction angle; γ unit weight (kN/m³); β slope angle; H soil thickness normal to the slope (m), $z = H/\cos\beta$ vertical depth; $m = h_w/z$ groundwater ratio; r_u pore-pressure ratio; k_h horizontal seismic coefficient; c_r root cohesion; K_s saturated hydraulic conductivity; P cumulative event rainfall; FS factor of safety; P_f probability of failure; β_R reliability index; S_i, $S_{T,i}$ first- and total-order Sobol' indices; HI, VI, RI hazard, vulnerability and risk indices.

1 Conceptualisation

1.1 Problem and objective

AHP-based landslide maps assign a weight to each factor through pairwise expert comparisons. Three properties of that approach limit its use for risk decisions: the weights depend on the panel that is consulted, a consistency ratio below 0.10 shows only that the judgements are coherent rather than correct, and the resulting index is not a probability, so it cannot enter a quantitative risk assessment (Fell et al., 2008; Corominas et al., 2014). The objective of this work is a model in which every quantity that drives the decision is either measured, taken from a published distribution, or computed from mechanics, and whose accuracy is tested, honestly and with its limits stated, against observed slope behaviour.

1.2 Conceptual framework

The framework is organised in five stages (Figure 1). Field observations are recorded as Varnes-type classes (scores 1 to 5) or, where available, as measured parameters. A parameter model converts each class into a probability distribution. Limit-equilibrium physics converts sampled parameters into a factor of safety, and Monte Carlo simulation converts the distribution of FS into P_f. A learning stage, validated on real slopes, corrects what idealised physics cannot represent. The decision stage combines hazard with an independent vulnerability index.

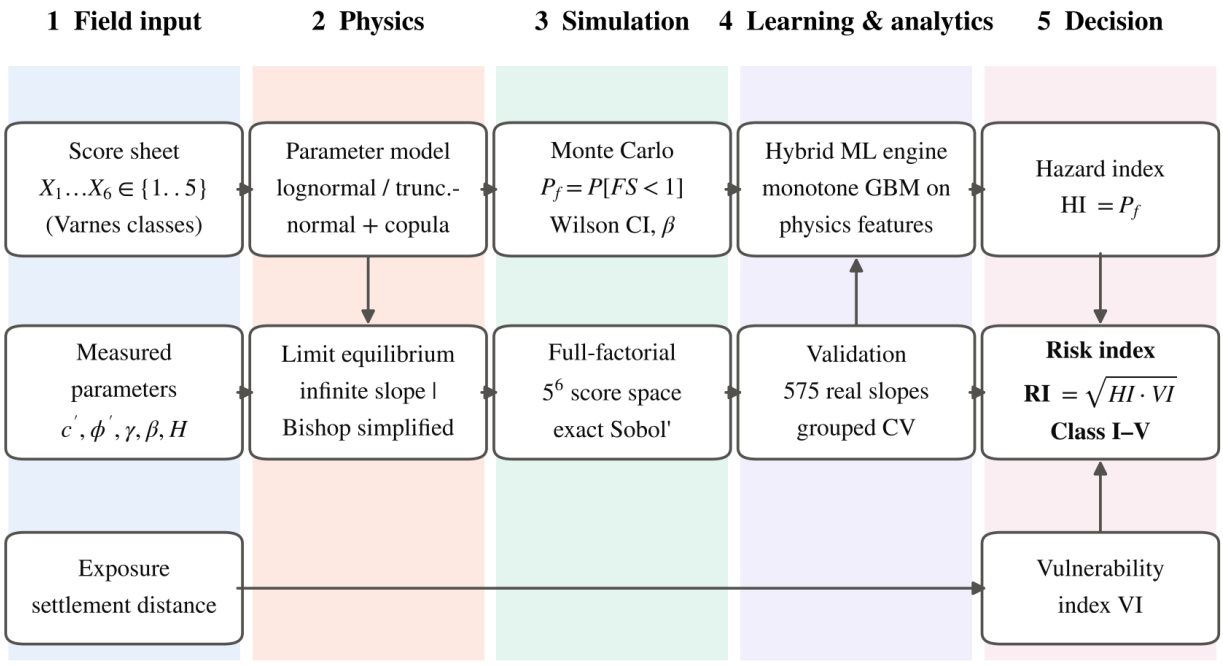


Figure 1. Conceptual framework. Judgement is confined to measurable class limits and published parameter distributions; weights, probabilities and classes are computed.

1.3 Parameter hierarchy

The model is a five-level hierarchy (Figure 2). The risk index (level 0) combines hazard and vulnerability (level 1). Hazard follows the Varnes distinction between preparatory factors that bring a slope towards its limit and triggering factors that cross it, with groundwater bridging both (level 2). Each of the six field variables (level 3) is resolved into the physical parameters that enter the FS equation (level 4). Rule 1 of the framework is that every hazard variable must change FS through an explicit physical term; variables that do not (for example distance to settlements) belong to vulnerability.

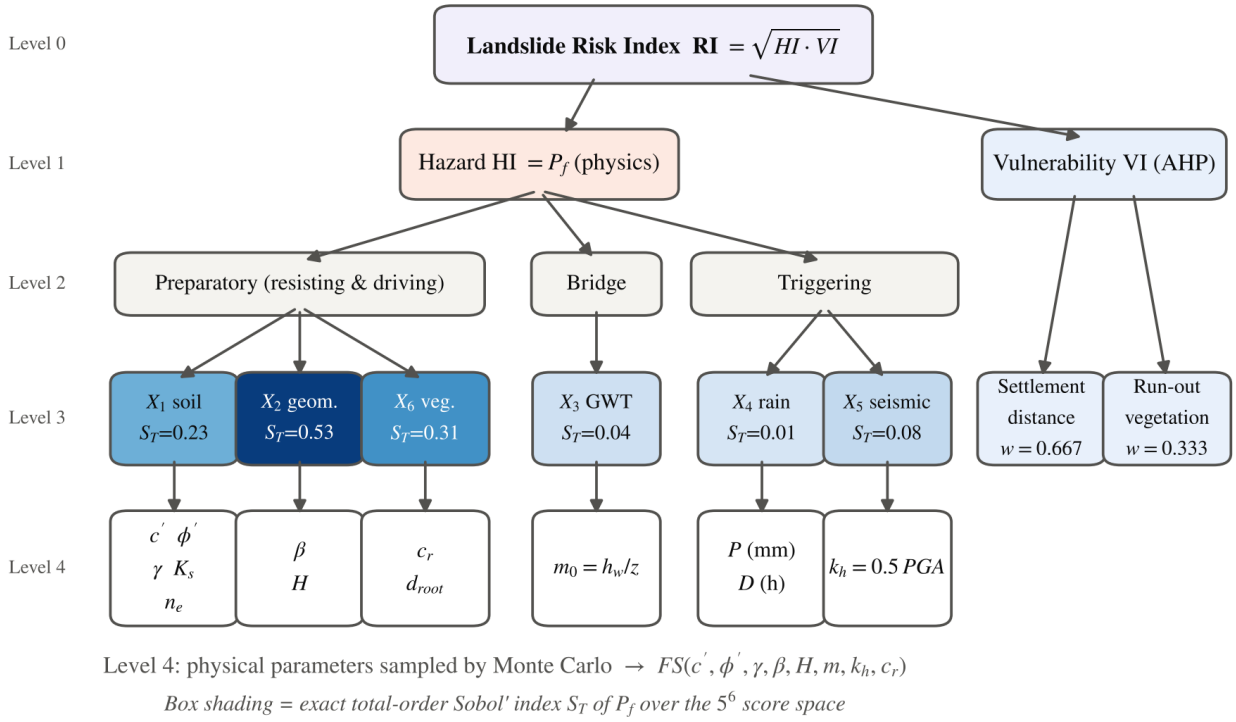


Figure 2. Parameter hierarchy from the risk index to the sampled physical parameters. Level-3 boxes are shaded by their exact total-order Sobol' index S_T of P_f (Section 3.5).

Table 1. Hazard variables, the physical parameters they control, and the probability model used inside each class.

Variable	Physical parameters	Distribution within a class	Source of class values
X1 Soil type	c', φ', γ, K_s, n_e	c' lognormal (COV 0.45–0.60); φ' normal truncated ±3σ (COV 0.08–0.20); γ normal (COV 0.05); K_s lognormal ($\sigma_{\log 10}$ 0.5); ρ(c',φ') = −0.5 via Gaussian copula	ISSMGE TC304 (2021); Phoon & Kulhawy (1999); Indonesian residual-soil tests
X2 Geometry	β, H	β uniform in class (10–60°); H uniform 1–3 m	Infinite-slope validity (L/H ≥ 25; Milledge et al., 2012)
X3 Groundwater	$m_0 = h_w/z$	uniform in class (0–1)	Field / piezometer class
X4 Rainfall	P, D	P uniform in class (0–600 mm); D uniform 24–72 h	Project assumption (class limits not yet tied to a cited threshold)
X5 Seismic	k_h	uniform in class (0–0.30)	k_h = 0.5·PGA (Hynes-Griffin & Franklin 1984)
X6 Vegetation	c_r, root depth	uniform in class (0–25 kPa; 0–3 m)	Stevens et al. (2006); Schmidt et al. (2001). Natural forest (25.6–94.3 kPa) excluded

2 Mathematical and physical formulation

2.1 Infinite slope (shallow translational failure)

For a soil layer of thickness H measured normal to the slope, the vertical depth is z = H/cos β. Resolving the forces on a vertical slice (Figure 3a) with pore pressure, pseudo-static seismic inertia and root reinforcement gives the factor of safety:

$$FS = \frac{c' + c_{r,\mathrm{eff}} + \left[\gamma z \cos^2\beta\,(1 - k_h \tan\beta) - \gamma_w m z \cos^2\beta\right]\tan\phi'}{\gamma z \cos\beta\,(\sin\beta + k_h \cos\beta)} \tag{1}$$

Failure occurs when FS < 1. For c' = 0, dry and static conditions the expression reduces to FS = tan φ'/tan β, which is used as a verification case. Measuring depth normal to the slope is essential: with a vertical depth held constant, the cohesion term c'/(γ z sin β cos β) passes through a minimum at 45° and FS spuriously increases on steeper slopes.

2.2 Rainfall, groundwater and roots

Rainfall acts through the groundwater ratio m. During an event of cumulative depth P and duration D, infiltration is limited by the saturated conductivity; the excess becomes runoff:

$$m = \min\left(1,\; m_0 + \frac{\min(P,\; K_s D)}{n_e\, z}\right) \tag{2}$$

Roots reinforce the slip plane only where they cross it:

$$c_{r,\mathrm{eff}} = c_r \min\left(1,\; d_{\mathrm{root}}/H\right), \qquad k_h = 0.5\,\mathrm{PGA}/g \tag{3}$$

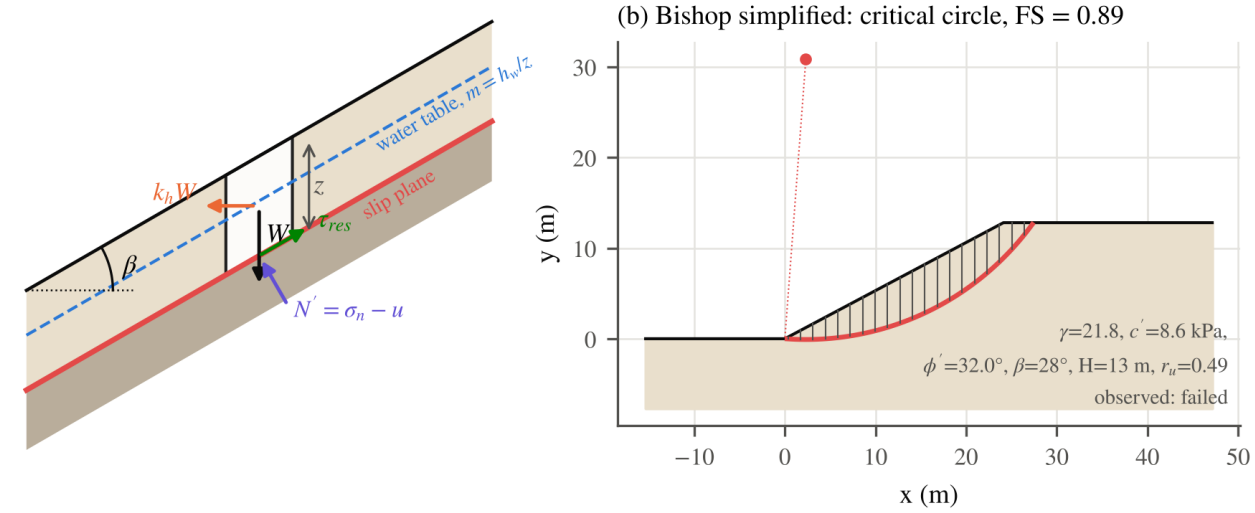


Figure 3. (a) Forces on a vertical slice of an infinite slope. (b) Critical slip circle and slices found by the Bishop simplified search for a failed case history (FS = 0.89).

2.3 Bishop simplified method (deep circular failure)

High slopes in the case-history database fail on circular surfaces, for which the infinite-slope model does not apply. The Bishop simplified method satisfies moment equilibrium about the circle centre with horizontal inter-slice forces, and is solved by fixed-point iteration:

$$FS = \frac{\sum_i \left[c' b_i + (W_i - u_i b_i)\tan\phi'\right]/m_{\alpha,i}}{\sum_i W_i \sin\alpha_i}, \qquad m_{\alpha,i} = \cos\alpha_i + \frac{\sin\alpha_i \tan\phi'}{FS}, \qquad u_i = r_u \gamma h_i \tag{4}$$

The critical circle is found by a vectorised search over 1,344 candidate circles defined by an exit point, an entry point and the sagitta of the arc. Two admissibility rules are enforced: both arc ends must lie on the lower half of the circle, and circles with $m_\alpha < 0.2$ on any loaded slice are rejected (a common practice; its attribution to Whitman & Bailey

(1967) could not be confirmed from an accessible source). Without these rules the search returned FS values up to 45% below Taylor's exact solution.

2.4 Probabilistic model

Within a soil class, c' and φ' are correlated through a Gaussian copula. With independent standard normal variables Z_1, Z_2 and correlation $\rho = -0.5$:

$$c' = F_{\mathrm{LN}}^{-1}(\Phi(Z_1)), \qquad \phi' = F_{\mathrm{TN}}^{-1}\left(\Phi(\rho Z_1 + \sqrt{1-\rho^2}\, Z_2)\right) \tag{5}$$

Negative c'–φ' correlation matters: with $\rho = 0$ the failure probability of a reference slope is about 1.05%, against 0.05% with $\rho = -0.5$ (Hammah et al., 2004; homogeneous slope, mean FS 1.21, normal c' and φ' with COV 10%, $\pm 3\sigma$ truncation). Truncation is not neutral: in the same study $\rho = -0.5$ gives 0.12% at $\pm 5\sigma$ but 0.05% at $\pm 3\sigma$, so truncating φ' at $\pm 3\sigma$ removes non-physical tails at the cost of lowering small P_f values.

2.5 Reliability measures

$$\hat{P}_f = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}[FS(\boldsymbol{\theta}_k) < 1], \qquad \mathrm{COV}(\hat{P}_f) = \sqrt{\frac{1-P_f}{N P_f}}, \qquad \beta_R = -\Phi^{-1}(P_f) \tag{6}$$

Confidence intervals use the Wilson score interval, which stays valid when no failures are observed. As an independent check, the First Order Reliability Method (FORM) finds the design point u* closest to the origin of standard normal space on the limit state $g(u) = FS - 1 = 0$, giving $\beta_R = |u^*|$ (Hasofer & Lind, 1974).

2.6 Variance-based sensitivity and risk index

Because the complete score space contains only $5^6 = 15{,}625$ combinations, all of them are simulated and Sobol' indices are computed exactly by analysis of variance rather than estimated:

$$S_i = \frac{\mathrm{Var}(E[P_f \mid X_i])}{\mathrm{Var}(P_f)}, \qquad S_{ij} = \frac{\mathrm{Var}(E[P_f \mid X_i, X_j]) - V_i - V_j}{\mathrm{Var}(P_f)}, \qquad S_{T,\,i} = 1 - \frac{\mathrm{Var}(E[P_f \mid X_{\sim i}])}{\mathrm{Var}(P_f)} \tag{7}$$

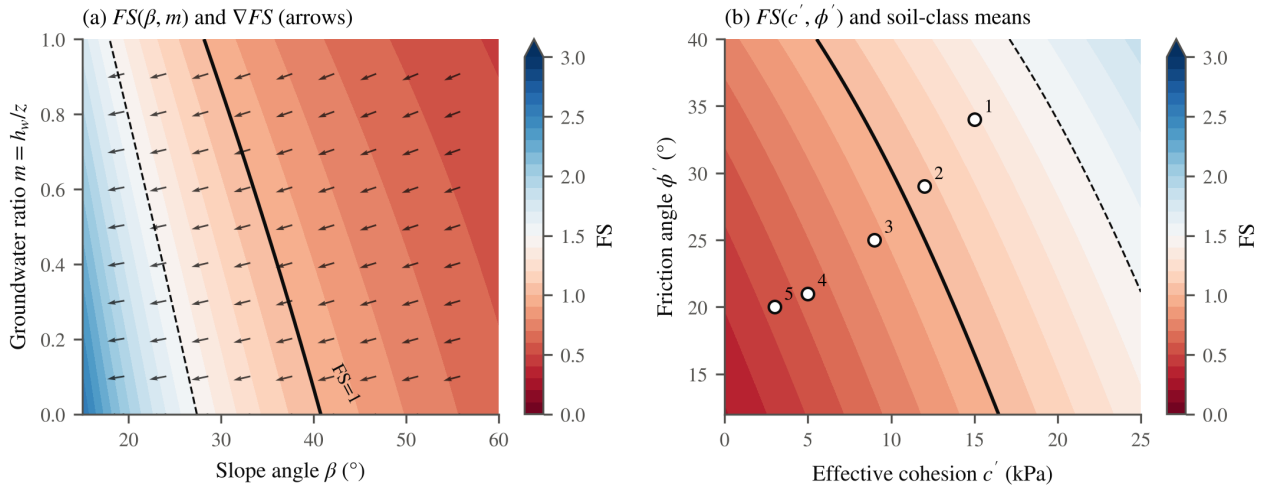
$$HI = P_f, \qquad VI = \frac{0.667\, s_{\mathrm{dist}} + 0.333\, s_{\mathrm{veg}} - 1}{4}, \qquad RI = \sqrt{HI \cdot VI} \tag{8}$$

The geometric mean ensures that risk vanishes when either hazard or vulnerability vanishes. Classes are $\mathrm{I} < 0.20 \le \mathrm{II} < 0.40 \le \mathrm{III} < 0.60 \le \mathrm{IV} < 0.80 \le \mathrm{V}$.

3 Model behaviour: gradients, matrices and parameter hierarchy

3.1 Response surfaces and the gradient field

Figure 4 shows FS over two parameter planes at a reference medium-clay state. The arrows give the direction of steepest increase of FS (∇FS) in axis-normalised coordinates. In the (β, m) plane the gradient points almost entirely towards lower slope angle: across the plotted ranges, varying β from 15° to 60° changes FS about 3.3 times more than varying m from 0 to 1 ($\partial FS/\partial\beta \approx -0.022$ per degree, $\partial FS/\partial m \approx -0.30$). The FS = 1 contour in the (c', φ') plane separates soil classes 1–2 (stable at β = 40°, m = 0.5) from classes 3–5.



Reference: γ=17 kN/m³, H=2 m, k_h=0.05, c_r=3 kPa; (a) c'=9 kPa, ϕ'=25°; (b) β=40°, m=0.5. Solid: FS=1, dashed: FS=1.5 (SNI 8460).

Figure 4. Factor of safety response surfaces. (a) Slope angle versus groundwater ratio, with the gradient field. (b) Cohesion versus friction angle, with the mean of each soil class.

3.2 Gradients and the sensitivity (Jacobian) matrix

Local sensitivity is expressed as the change in FS caused by a one-standard-deviation change of each parameter, $\sigma_j \partial FS/\partial x_j$, so that parameters with different units are comparable. Figure 5a shows that cohesion is the dominant stabilising parameter at every slope angle, that its effect decreases as slopes steepen, and that the destabilising effect of slope angle is strongest on gentle slopes. Figure 5b assembles these gradients into a Jacobian matrix for each soil class: cohesion and soil thickness H dominate, and the magnitude of every term shrinks from class 1 to class 5 because weak soils are already near failure.

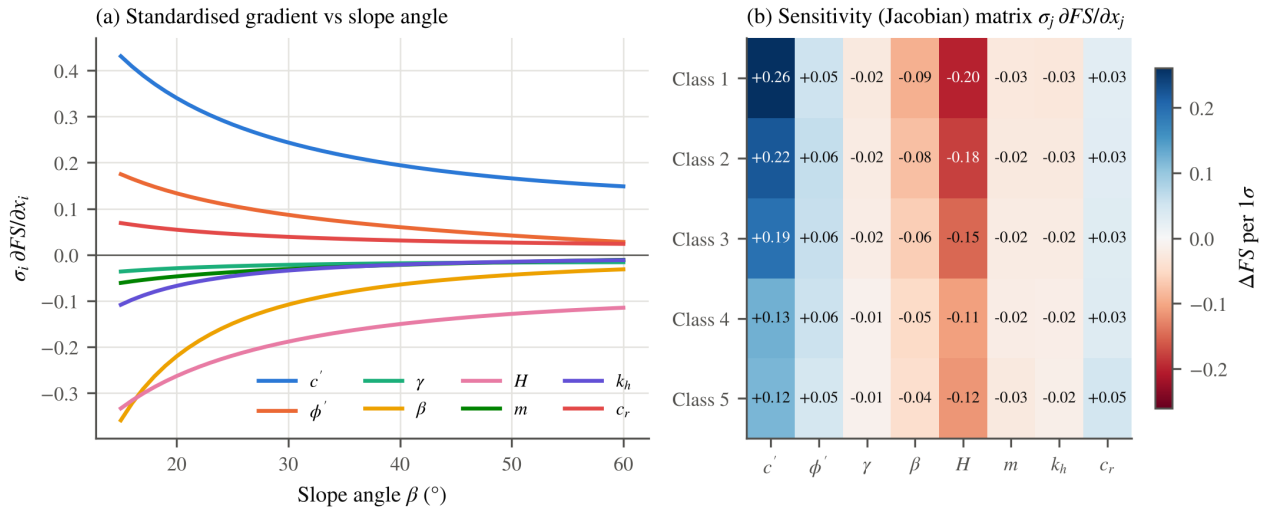


Figure 5. (a) Standardised gradient of FS with respect to each parameter as a function of slope angle. (b) Jacobian matrix of standardised gradients for the five soil classes (β = 40°).

3.3 Matrices between parameters

Three matrices describe how parameters relate to each other and to the output (Figure 6). The Spearman rank correlation matrix of sampled inputs and FS (6a) shows the imposed c'–φ' correlation (−0.5) and confirms that c' is the input most strongly associated with FS (ρ = 0.6) in a medium-clay slope. The Hessian matrix $\sigma_i\sigma_j\,\partial^2 FS/\partial x_i \partial x_j$ (6b) measures local interaction: the largest off-diagonal term couples c' and H, because cohesion is divided by the weight of the soil column. The second-order Sobol' matrix (6c) measures global interaction over the score space; the strongest pair is soil × geometry (S_{12} = 0.035), consistent with the slide-deck finding that weak soils fail when combined with steep geometry.

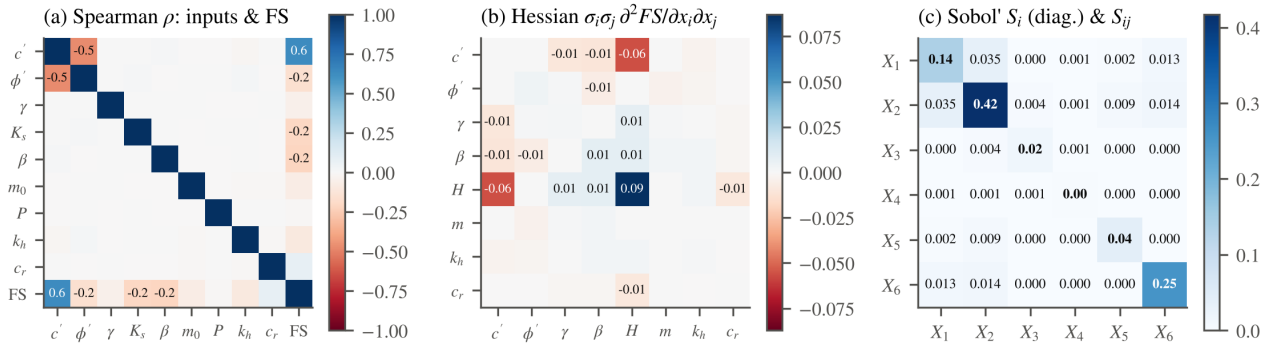


Figure 6. (a) Spearman correlation of sampled parameters and FS (all scores = 3). (b) Standardised Hessian of FS at the reference state. (c) Exact first-order (diagonal) and second-order (off-diagonal) Sobol' indices of P_f over the 15,625 score combinations.

3.4 Monte Carlo behaviour

For the case-study slope (medium clay, 45° slope, shallow groundwater, high rainfall, static, coffee plantation) the simulated FS distribution is right-skewed with P_f = 0.860 (95% CI 0.855–0.864, N = 20,000). The running estimate stabilises after a few thousand samples (Figure 7b). The 400 samples per configuration used for the score table bound the 95% confidence half-width at 0.049; for rare failures far larger samples are needed (Figure 7c), e.g. about 10^5 for $P_f = 10^{-3}$ at COV 0.10.

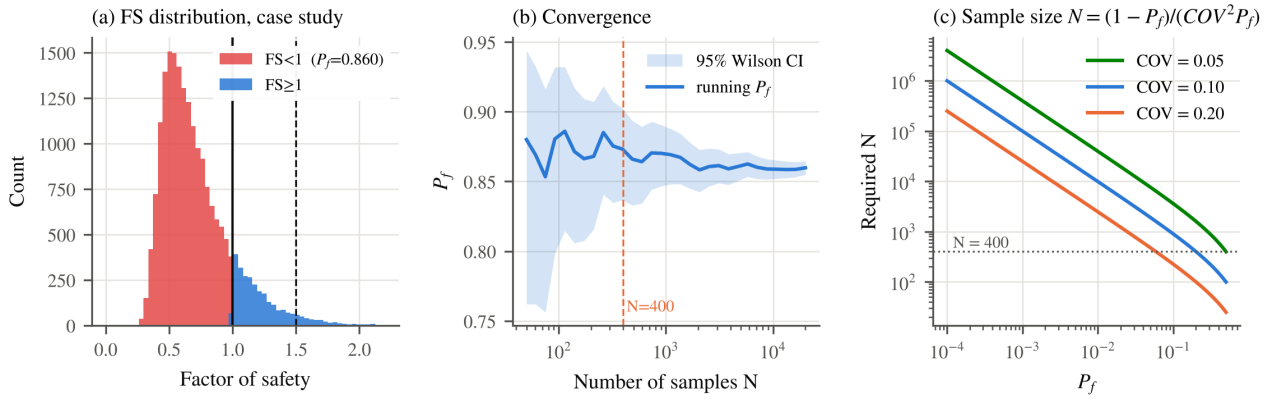


Figure 7. (a) FS distribution for the case-study slope. (b) Convergence of P_f with its 95% Wilson interval. (c) Sample size required for a target coefficient of variation of the P_f estimator.

3.5 Hierarchy of parameters by variance decomposition

Under the v2 class definitions, the exact Sobol' indices rank the field variables as follows (Table 2, Figure 8); Section 7.3 shows how this ranking changes under other class definitions. Slope geometry is the dominant driver, followed by vegetation and soil type; together they explain over 85% of the variance of P_f across the score space. Groundwater, seismic load and rainfall have small first-order effects, and their total effects are 1.8 to 4.7 times larger than their first-order effects, i.e. they act mainly through interactions. Rainfall now has a non-zero, monotone effect because cumulative event rainfall replaced hourly intensity; its share remains small because the model is static (Section 7).

Table 2. Exact Sobol' indices of P_f (model v2, 15,625 combinations × 400 samples). Differences between three random seeds are below 0.001.

Rank	Variable	S_i (first order)	$S_{T,i}$ (total)	Interaction share $(S_T - S_i)/S_T$	Normalised weight $S_T/\Sigma S_T$
1	Geometry	0.417	0.527	21%	44.1%
2	Vegetation	0.255	0.313	19%	26.2%
3	Soil	0.137	0.227	40%	19.0%
4	Seismic	0.042	0.076	44%	6.4%
5	Groundwater	0.016	0.038	58%	3.2%
6	Rainfall	0.003	0.014	78%	1.1%

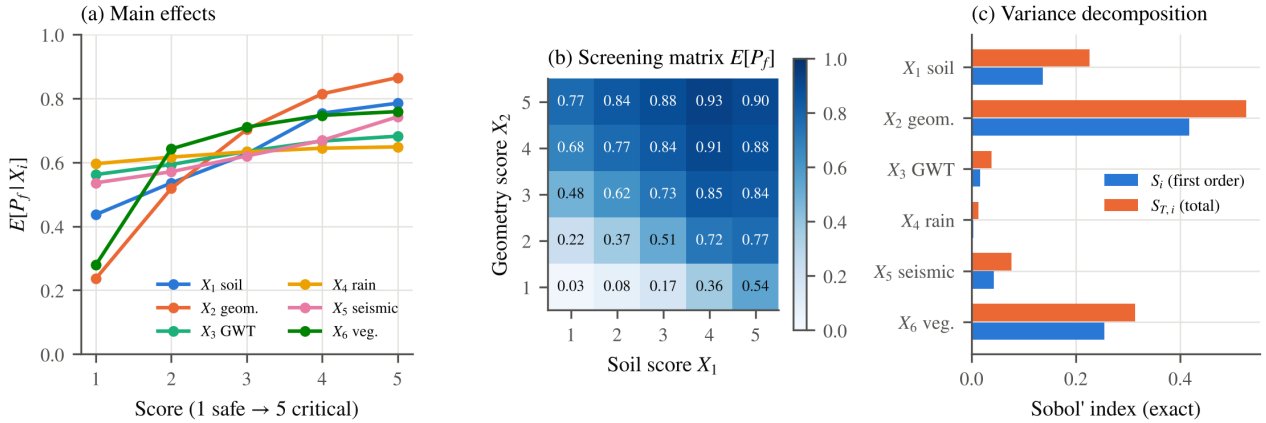


Figure 8. (a) Main effects $E[P_f \mid X_i]$. (b) Soil × geometry screening matrix. (c) First-order and total Sobol' indices.

4 The machine engine

4.1 Architecture

The machine engine combines the physics solver with a learning model (Figure 9). For each slope the Bishop solver computes FS, from which three physics features are derived: ln FS, the stability number $N_s = c'/\gamma H$ and the friction ratio tan φ'/tan β. These are combined with the six raw inputs in a gradient-boosting classifier with *monotone constraints* on every input that has a definite physical direction: the predicted probability of failure may not decrease when β, r_u, γ or H increase, and may not increase when FS, c', φ', N_s or the friction ratio increase (for $c' \geq 0$, Bishop FS never rises with γ or H). An earlier version left γ and H unconstrained; the audit showed that it then learned jagged, non-physical responses to these two inputs that raised AUC by 0.02–0.04 (Section 7).

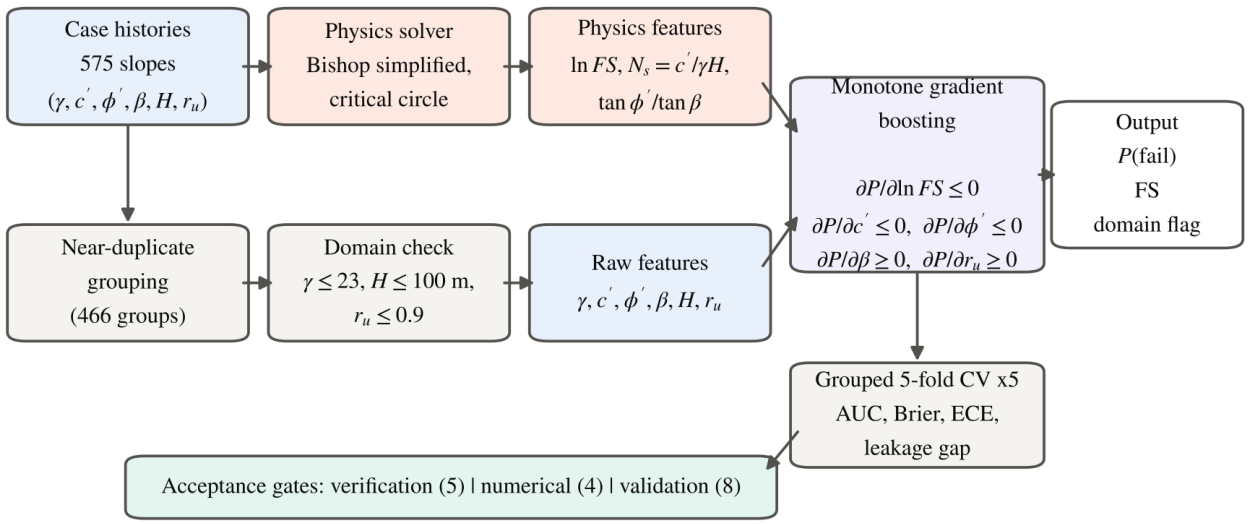


Figure 9. Architecture of the hybrid physics–machine-learning engine and its validation loop.

4.2 Data

The engine was trained and tested on the open database of Ma et al. (2022), 627 slope case histories with γ, c', φ', β, H, r_u and observed status. 52 rows with physically impossible unit weights (γ < 10 kN/m³) were removed (56% of them failures), leaving 575 cases. The compilations overlap, so near-duplicate cases were assigned to 466 groups that are never split between training and test folds; 10 groups contain the same slope labelled both stable and failed. The database is strikingly inconsistent with homogeneous slope mechanics: 150 of 288 'stable' slopes have a Bishop FS below 1 and 78 of 287 'failed' slopes have FS ≥ 1. Many entries are high rock or mine slopes, and recorded strengths are often back-analysed; the data therefore test a classifier of reported outcomes, not the infinite-slope field model.

4.3 Models compared and validation design

Five models were evaluated on identical folds: (A) an equal-weight score index, each input scored by quintile with an assumed direction (this is *not* AHP: no expert weights were elicited or tested); (B) physics only, a two-parameter logistic calibration of ln FS; (E) logistic regression on the six raw inputs, the fair linear baseline that represents the best any fixed weighting of these inputs can do; (C) unconstrained gradient boosting; and (D) the fully monotone hybrid. Validation used stratified group 5-fold cross-validation repeated five times. The **primary estimator** pools the out-of-fold predictions of each repeat, in which every case is predicted once by a model that never saw it, and averages the metrics over repeats. The **primary population** is all cases; the soil-slope domain (γ ≤ 23 kN/m³, H ≤ 100 m, $r_u \leq 0.9$) was defined after seeing early results and is reported only as a secondary analysis.

Table 3. Out-of-fold performance, grouped cross-validation, all cases (n = 575); single-pass estimator averaged over 5 repeats; threshold 0.5. The last column is the post-hoc soil domain (n = 327).

Model	AUC	Accuracy	Sensitivity	Specificity	Brier	ECE	AUC soil domain
A Equal-weight score index (not AHP)	0.581	0.568	0.594	0.542	0.247	0.074	0.680
B Physics only (Bishop FS)	0.645	0.567	0.474	0.660	0.237	0.054	0.756
E Logistic regression, raw inputs	0.731	0.665	0.631	0.698	0.212	0.063	0.820
C Gradient boosting, unconstrained	0.906	0.837	0.845	0.828	0.123	0.056	0.910
D Hybrid, fully physics-monotone	0.860	0.801	0.808	0.795	0.151	0.077	0.870

4.4 Results

The hybrid reaches AUC 0.860 (group-bootstrap 95% CI 0.832–0.896). It exceeds logistic regression by 0.132 (CI 0.093–0.172) and physics alone by a wider margin. The unconstrained learner C scores higher still, but part of that advantage comes from non-physical patterns in γ and H; the constrained hybrid is preferred because every response it learns is admissible. At the default threshold the hybrid misses 19% of failed slopes. A threshold chosen to catch 90% of failures (0.27 on the averaged predictions) keeps specificity at 69%; for life-safety screening such a recall-oriented threshold is more appropriate than 0.5.

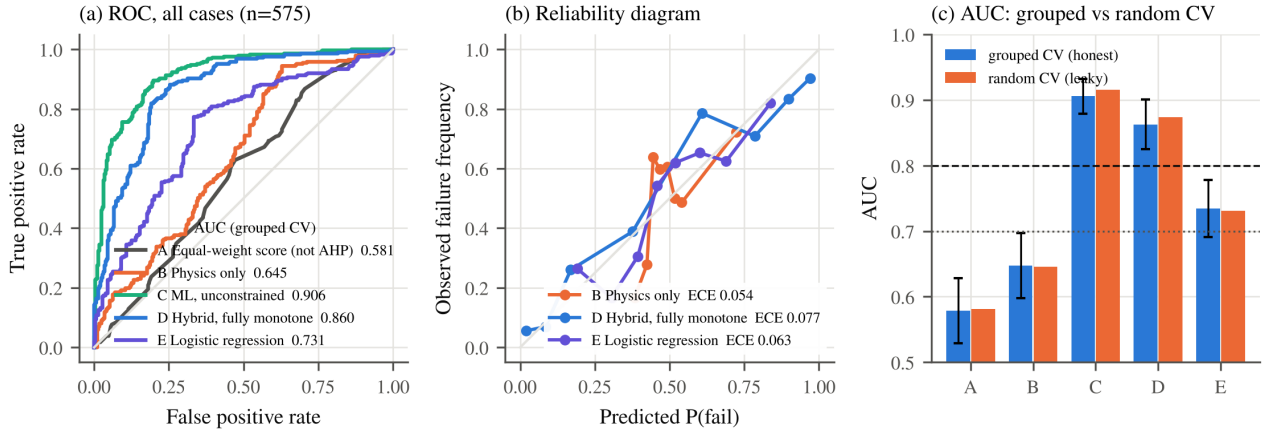


Figure 10. (a) ROC curves of the five models (pooled out-of-fold, all cases). (b) Reliability diagram. (c) Mean fold AUC (± 1 s.d.) under grouped and random cross-validation; dashed lines mark the 0.80 target and 0.70 minimum.

4.5 How the machine models

Figure 11 opens the model. Permutation importance (11a) measures the loss of AUC when one feature is shuffled; partial-dependence and individual conditional expectation curves (11b) show the learned response to each input. Every response is monotone in the physically required direction, including γ and H. When the three physics features are shuffled together the AUC drops by 0.030, against 0.362 for the raw block: the learner extracts most of its skill from the raw parameters and uses physics mainly as structure and constraint.

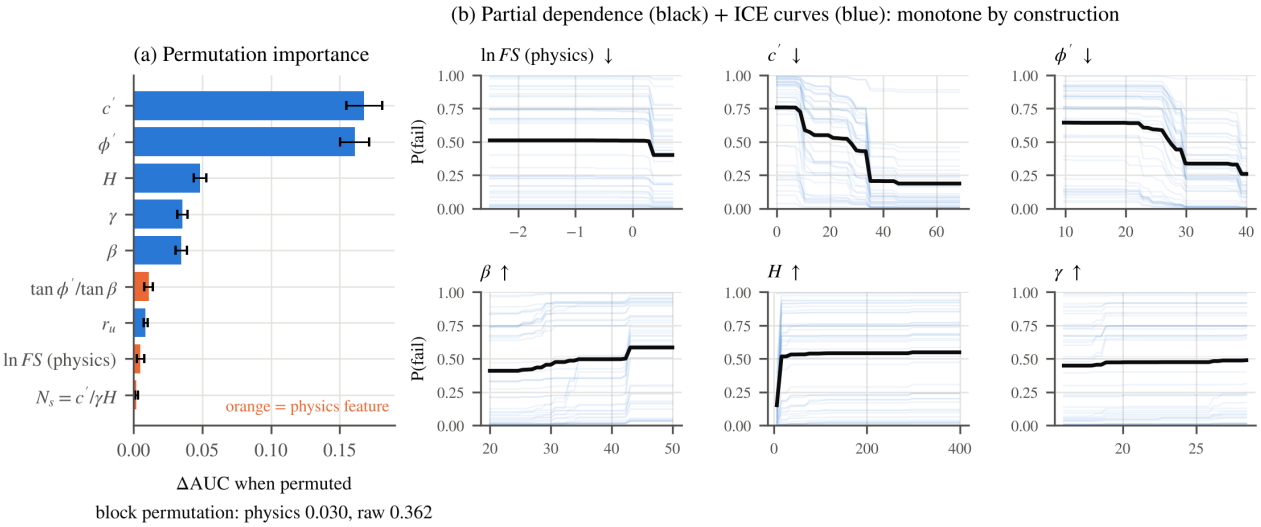


Figure 11. (a) Permutation importance (ΔAUC) of the hybrid model; orange bars are physics features. (b) Partial dependence (black) and ICE curves (blue); arrows give the imposed monotone direction.

5 Accuracy, acceptance and reliability

Acceptance follows the verification and validation (V&V) framework of Oberkampf, Trucano & Hirsch (2003). **Verification** asks whether equations are solved correctly; **numerical** checks ask whether simulation noise could change a conclusion; **validation** asks whether predictions match observed behaviour. Each threshold is labelled by its basis: *literature* (taken from a cited source), *project-set* (chosen by this project, not an external standard) or *analytic* (true by design, so a design check rather than a test). Of the 16 passes, 4 are analytic. The single failure is substantive: physics-derived P_f does not behave as a frequency on real slopes. The two pending criteria concern the tool that would actually be used in the field.

Table 4. Acceptance protocol (audited). 16 pass, 1 fail, 2 pending; shaded rows are not passed.

Gate	Code	Criterion	Value	Threshold	Basis	Status
Verif.	V1	Infinite slope, c'=0 dry: FS = tan(phi)/tan(beta)	1.11e-16	abs err < 1e-9	analytic	PASS
Verif.	V2	Bishop vs Taylor chart, phi=0, beta 45/60/75/90 deg (errors 3.7%, 0.1%, 0.2%, 0.1%)	0.037	max rel err <= 5%	literature	PASS
Verif.	V3	Bishop, c'=0: converges to tan(phi)/tan(beta)	0.001	rel err <= 2%	analytic	PASS
Verif.	V4	Bishop vs ACADS 1(a) (reference FS ~1.00, recalled, unverified)	0.009	rel err <= 3%	literature	PASS
Verif.	V5	FORM vs Monte Carlo (one test case, Pf ~0.97)	7.97e-04	abs diff Pf <= 0.01	project-set	PASS
Verif.	V6	Monotone constraints implemented: +10% c' never raises P(fail)	0.000	= 0	analytic	PASS
Numer.	N1	Max 95% CI half-width of Pf per configuration (N=400)	0.049	<= 0.05	analytic	PASS
Numer.	N2	Seed stability: mean |dPf| between seeds	0.016	<= 0.02	project-set	PASS
Numer.	N3	Seed stability: max |dS_T| of Sobol indices	3.57e-04	<= 0.02	project-set	PASS
Numer.	N4	Site Pf estimator COV (case study, N=20,000)	0.003	<= 0.10	literature	PASS
Valid.	A1	Hybrid AUC, grouped CV, ALL cases (group-bootstrap 95% CI lower bound)	0.832	>= 0.70 and point >= 0.80	literature	PASS
Valid.	A2	Hybrid beats logistic regression on raw inputs (dAUC, CI lower bound)	0.093	> 0	project-set	PASS
Valid.	A3	Hybrid beats physics-only FS (dAUC, CI lower bound)	0.175	> 0	project-set	PASS
Valid.	A4	Hybrid calibration: ECE on single-pass OOF predictions	0.077	<= 0.10	project-set	PASS
Valid.	A5	Hybrid Brier score (no-skill = 0.25)	0.151	<= 0.20	project-set	PASS
Valid.	A6	Leakage gap: AUC(random CV) - AUC(grouped CV)	0.014	<= 0.05	project-set	PASS
Valid.	A7	Physics Monte Carlo Pf is a calibrated probability (ECE of raw Pf, no fitting)	0.331	<= 0.10	project-set	FAIL
Valid.	A8	Score-based HI (6 field scores -> Pf) validated against observed landslides	–	AUC >= 0.80 and ECE <= 0.10	literature	PENDING
Valid.	A9	Spatial/temporal validation on an Indonesian landslide inventory	–	AUC >= 0.80 on spatial blocks	literature	PENDING

Site-level reliability. For decisions on individual slopes a dual criterion is recommended: the SNI 8460:2017 minimum factor of safety *and* a reliability index $\beta_R \geq 3$. For soil slopes SNI 8460:2017 (Tabel 25) requires FS 1.25–1.5 when uncertainty is low and 1.5–≥2.0 when it is high, depending on the cost and consequence of failure; the 1.5 permanent / 1.3 temporary values apply to rock slopes (Tabel 26); pseudostatic analysis requires FS > 1.1 (§7.5.1.1), with $k_h = 0.5 \cdot PGA$ (§12.2.7). A mean FS of 1.5 corresponds (for lognormal FS) to $\beta_R \approx 4.0$ when the COV of FS is 0.10 but only ≈ 2.0 when it is 0.20. Because criterion A7 fails, β_R computed from literature COVs should be treated as a relative measure until it is calibrated against local outcomes.

6 Risk index output and case study

The case study from the original slide decks (medium clay, 45° slope, shallow groundwater, high rainfall, static, coffee plantation, settlement 80 m away) is reproduced exactly by the weighted-sum method (HI = 0.575, VI = 0.584, RI = 0.579, Class III). With the physics-based probability, HI = P_f = 0.86 and RI = 0.71, i.e. Class IV. This shift is robust to the Monte Carlo assumptions (P_f 0.85–0.93 across seven variants) and to moderate scoring error (P(Class IV) = 0.95 if each score has a 10% chance of being one class off). It is nonetheless a model output, not a validated finding: the score-based index has not been tested against observed landslides (criterion A8), and the RI class limits were originally defined for the weighted score index, not for a probability, so mixing the two scales is itself an unvalidated choice.

Table 5. Case-study risk index under the three hazard formulations (model outputs, not validated).

Method	HI	VI	RI	Class
Weighted sum, slide weights	0.575	0.584	0.579	III Medium
HI = P_f (model v1, uniform ranges)	0.848	0.584	0.703	IV High
HI = P_f (model v2, literature distributions)	0.860	0.584	0.709	IV High

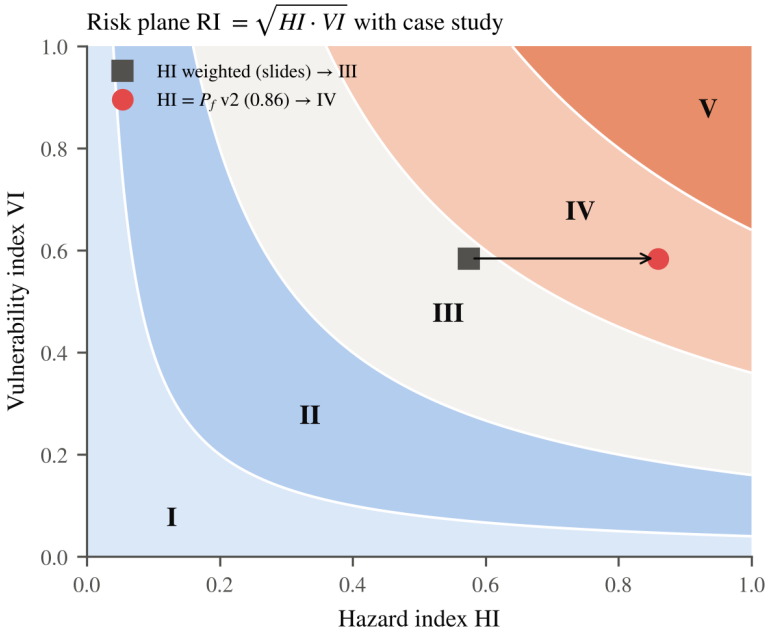


Figure 12. Risk plane RI = $\sqrt{(HI \cdot VI)}$ with class boundaries. The case study moves from Class III (weighted sum) to Class IV (physics-based P_f).

7 Audit: corrected claims, validation bias, the role of Monte Carlo and scoring

The analysis was audited after version 2.0 with one question: is every reported number an objective, reproducible estimate of how the model would perform on new slopes? Each suspected bias was measured rather than argued; all numbers below come from *audit/run_audit.py* and *results/audit/*.

7.1 Corrected claims

Table 6. Claims corrected by the audit.

Earlier claim (v2.0 or chat summary)	Problem found	Corrected statement
Hybrid AUC 0.903 (soil domain) / 0.906	Post-hoc domain; predictions averaged over 5 repeats (ensemble, +0.007); γ and H unconstrained (+0.02–0.04, non-physical)	AUC 0.860 on all cases with a fully constrained model
Hybrid 'clearly beats an AHP-style index'	Baseline A was an equal-weight score designed here; no AHP weights were tested	Hybrid beats a fair logistic baseline by 0.13 AUC; comparison with AHP is untested
'16 of 17 acceptance criteria pass'	Four passes are true by design; most thresholds were set by this project; the calibration estimator was changed after the per-fold estimate failed	16 of 19 pass (4 analytic), 1 fail, 2 pending; thresholds labelled by basis
The engine is validated	Validation used a global case database with different inputs from the field score sheet	Only the Bishop-hybrid classifier is validated; the score-based HI/RI is not (A8)
P_f is a probability usable in risk assessment	Raw Monte Carlo P_f on real slopes: ECE 0.33, worse than a constant guess	P_f ranks slopes; it needs empirical calibration before use as a frequency
Sobol' weights are objective and replace AHP weights	Normalised weights move with class definitions: soil 0.18–0.37, vegetation 0.07–0.28	Judgement moved from weights to class tables; it was reduced and made explicit, not removed
Bishop matches Taylor within 0.2%	The 45° toe-circle case (3.7% error) had been left out; at 45° the global critical circle is a deep base circle (Ns ≈ 0.181)	0.1–0.2% for 60–90°, 3.7% at 45°
Case-history AUC compared with published map AUCs	Slope-classification AUC and susceptibility-map AUC measure different tasks	Not compared

7.2 Validation bias

Estimator choice. For the adopted hybrid, per-fold mean AUC is 0.863, the single-pass pooled estimate 0.860 and the 5-repeat averaged estimate 0.865; the last is an ensemble and slightly optimistic, and is no longer reported. Expected calibration error ranges from 0.071 to 0.116 depending on the estimator; per-fold ECE is inflated by small folds, averaged ECE deflated by ensembling, and the single-pass value is used. **Domain choice.** Across 48 alternative domain filters, physics-only AUC ranges 0.64–0.78 and the hybrid 0.84–0.91; the filter used earlier was not the most favourable, but physics-only results are clearly sensitive to it. **Leakage.** Random cross-validation overstates the hybrid AUC by 0.014. Grouping duplicates up to eight times more coarsely leaves the constrained hybrid within 0.858–0.868, so residual near-duplicate leakage is small for the adopted model. **Bootstrap.** Resampling whole duplicate groups gives confidence intervals almost identical to case-level resampling, so interval widths are not materially understated.

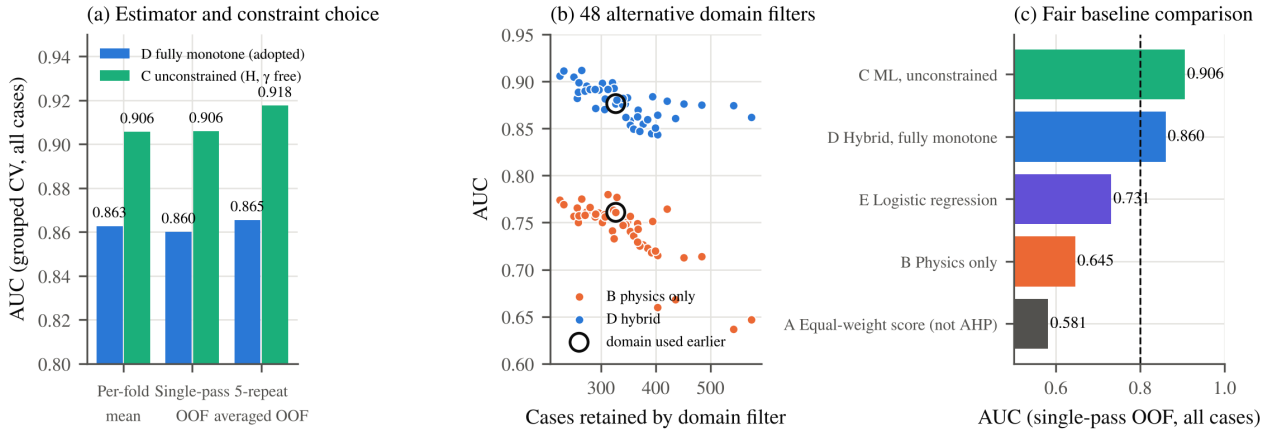


Figure 13. (a) Reported AUC depends on the estimator and on whether γ and H are constrained. (b) Physics-only and hybrid AUC under 48 alternative domain filters; circles mark the filter used in v2.0. (c) Fair comparison of all five models on all cases.

7.3 How Monte Carlo affects the model

Monte Carlo plays two different roles, and the audit separates them. **Inside the score model** it converts each class into a distribution of FS and hence P_f. At N = 400 its sampling noise is small (s.d. of the case-study P_f = 0.019) compared with the effect of a 20% chance of mis-scoring each variable (90% interval of P_f 0.41–0.97). The distributional choices that Monte Carlo makes explicit matter unevenly: c'–φ' correlation and COV move the normalised weights by about 0.02 or less, whereas the class definitions (v1 uniform ranges versus v2 literature distributions) and the root-cohesion range move them substantially (geometry 0.43–0.55, vegetation 0.07–0.28). **On real slopes** Monte Carlo P_f discriminates no better than the deterministic FS it is built on (AUC 0.633 versus 0.650), and replacing ln FS by P_f in the hybrid leaves AUC unchanged (0.863 versus 0.862). Its probabilities are poorly calibrated: slopes with $P_f \leq 0.1$ failed 22% of the time and slopes with $P_f > 0.9$ only 57% (Figure 14a). The value of Monte Carlo is therefore transparent uncertainty bookkeeping and a graded output, not additional predictive accuracy; its probabilities become meaningful only after calibration against outcomes.

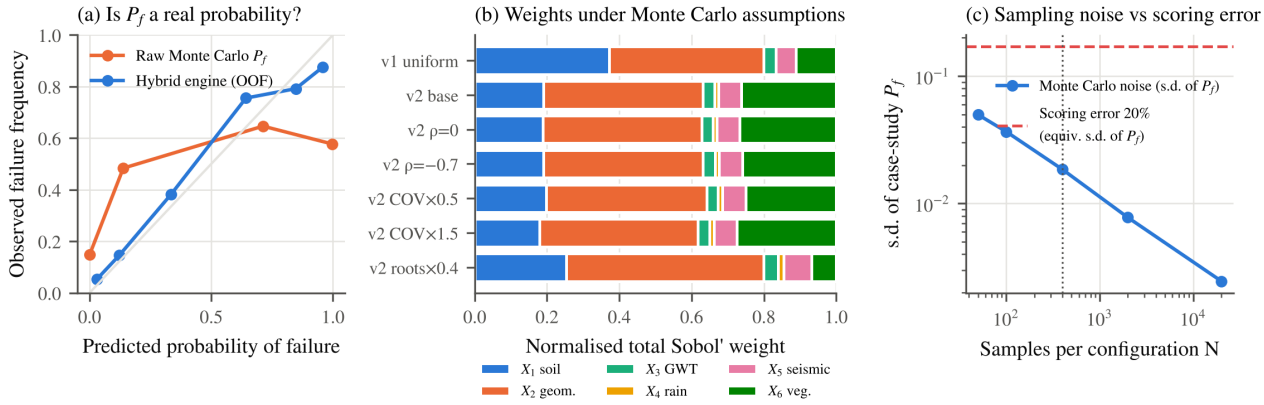


Figure 14. (a) Calibration of raw Monte Carlo P_f and of the hybrid engine on real slopes. (b) Normalised total Sobol' weights under seven Monte Carlo assumption sets. (c) Monte Carlo sampling noise versus the spread caused by a 20% scoring error.

7.4 How the score is taken and how scoring error propagates

A field engineer assigns each variable a class from a written table (Table 1): for example a 45° slope falls in geometry class 4 (45–55°). The six scores index one row of a pre-computed table of 15,625 P_f values, which becomes HI; the settlement distance and vegetation scores give VI, and RI = $\sqrt{(HI \cdot VI)}$ gives the class (Figure 15a). No weights are applied at this stage: the influence of each variable is already contained in the physics. Two consequences follow. First, the result is only as good as the scoring: with a 20% chance of each score being one class off, the case study remains Class IV with probability 0.87 but falls to Class III with probability 0.10 (Figure 15b). Second, the physics-based and weighted-sum hazards rank configurations similarly (Spearman ρ = 0.76) yet assign the same RI class in only 45% of the 15,625 combinations, with the physics version higher in 35%. Which of the two is closer to reality cannot be decided without outcome data (A8).

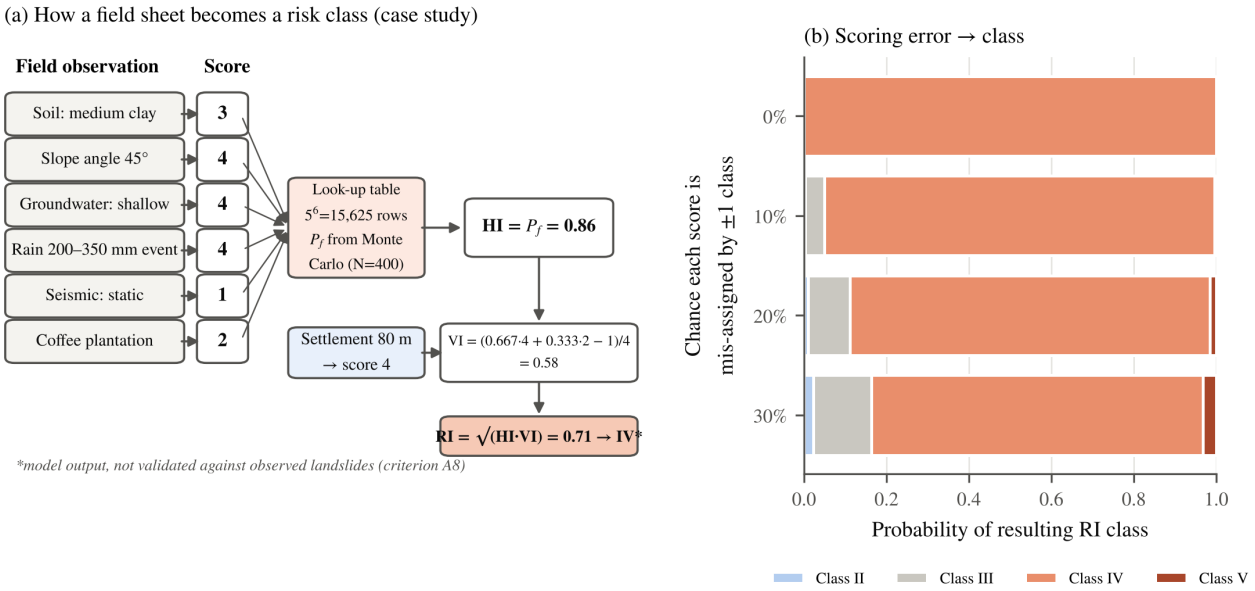


Figure 15. (a) Worked example: from field observations to scores, the P_f look-up table, HI, VI and the risk class. (b) Probability of each RI class for the case study when every score has a given chance of being mis-assigned by one class.

8 Position in the literature, limitations, ethical use and next steps

8.1 Position

Combining the infinite-slope equation with Monte Carlo sampling is established (SINMAP, TRIGRS-P, HIRESSS, Landlab). TRIGRS-P improved predictive power over deterministic TRIGRS by about 10% and 16% in two test areas, Frontignano (Italy) and Mukilteo (USA) (Raia et al., 2014). Published susceptibility maps report AUC values on landslide inventories; those values measure a different task from the slope-classification AUC in Section 4 and are not compared with it here. The literature search found no study that replaces AHP factor weights by variance-based indices computed from a physical model, nor a probabilistic infinite-slope or TRIGRS-P application in Indonesia; both are candidate contributions that still need confirmation by a systematic Scopus search.

8.2 Limitations

- **The field tool is unvalidated.** The score-based HI/RI has no outcome data yet (A8, A9). Validation in Section 4 concerns a different model with measured inputs, on a global database.
- **Probabilities are uncalibrated.** Physics-derived P_f is not a frequency on real slopes (A7 fails).
- **Static hydrology.** Transient infiltration, antecedent rainfall and loss of suction are not modelled, so the importance of rainfall is a lower bound.
- **Class definitions carry the judgement.** Class means for soils 2–5 and root-cohesion ranges are syntheses of published values; they dominate the variance-based weights.
- **Data quality.** The case database mixes soil, rock and mine slopes, contains conflicting labels and frequently disagrees with homogeneous slope mechanics.
- **Spatial structure and mechanism.** Parameters are sampled independently per slope; infinite slope applies only to shallow translational slides (length/depth ≥ 25; Milledge et al., 2012).

8.3 Ethical use

A landslide model influences decisions that affect lives and livelihoods, so its errors are not symmetric: a missed unstable slope can cost lives, while a false alarm costs money and trust. The following conditions are proposed for any use of this work:

- Until criteria A8 and A9 pass, outputs are for research and screening only and must not be the sole basis for evacuation, relocation or land-use restrictions; every Class IV–V result requires review by a geotechnical engineer.
- Report recall (the share of failures detected) alongside AUC, and choose decision thresholds by the cost of missed failures rather than by accuracy.
- Publish the class tables, distributions, code and data with every map, so that assumptions can be checked and results reproduced; state which numbers are validated, which are verified only, and which are assumptions.
- Do not present the variance-based weights as assumption-free; report their range under alternative class definitions.
- Respect data licences (Ma et al., 2022 is open access) and do not combine community-level risk classes with personal data without consent.

8.4 Next steps

- Spatial validation (A9) with the Samodra (2024) Cyclone Cempaka inventory and the BNPB InaRISK hazard index as the official baseline, using spatially blocked cross-validation; record field scores for mapped landslides to test A8.
- Calibrate P_f against outcomes (isotonic or Platt scaling on held-out data) before it is used as a probability.
- Calibrate class distributions with local laboratory data and SoilKsatDB; report weight ranges under $\pm 20\%$ class limits.
- Add transient hydrology (Iverson/TRIGRS) with 1–3-day antecedent rainfall.

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